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Using Machine learning to predict blood glucose level based on Photoplethysmography

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ABSTRACT

Photoplethysmography (PPG) is a non-invasive technique used to monitor the tiny changes in blood vessels caused by the heartbeat. This study obtained PPG signals from a commercial PPG module, followed by Signal Quality Index (SQI) detection to assess the signal quality and further signal processing, such as filtering and baseline shift. As a result, a total of 36 features were extracted from PPG signals in both time and frequency domains, and a correlation matrix was used to inspect the correlation among features. Machine learning models, including Random Forest (RF) and eXtreme Gradient Boosting (XGBoost), were employed to train and predict blood glucose levels. Of the 36 features, 9 were selected for use as input data for red and infrared light signals. Four datasets, namely “Red”, “Infrared”, “Composite” (a combination of “Red” and “Infrared”), and “Modified Composite” (a refined version of the “Composite” dataset that reduces feature collinearity and enhances prediction accuracy), were investigated. The RF model trained with the “Modified Composite” dataset yielded the best prediction, with a Mean Absolute Relative Difference (MARD) of 5.15% and an R-value of 0.93.

1. Introduction

Diabetes is a chronic disease characterized by abnormally high blood glucose levels in the body. Now, there is a lot of research on non-invasive (NI) blood glucose measurement. Many types of non-invasive (NI) methods are used to measure blood glucose, some of which use optical techniques such as polarized light [1,2], fiber optics [3], near-infrared spectrum [4,5], and Raman spectrum [3,6]. Chen et al. [1] proposed a method for determining the glucose concentration on the human fingertip by extracting two optical parameters, namely the optical rotation angle and the depolarization index, using a Mueller optical coherence tomography technique based upon polarizations and a genetic algorithm. This approach improves measurement precision and effectiveness. Koyama et al. [3] measured blood glucose levels using mid-infrared attenuated total reflectance (ATR) spectroscopy and hollow fiber optics. Also, Han et al. [4] used near-infrared spectroscopy to study the relationship with blood glucose concentration. Due to the differences in human tissue and physical fitness, the relationship between the data obtained by using near-infrared spectroscopy and blood glucose concentration is nonlinear, so a hybrid model combining partial least square regression (PLSR) and stacked auto-encoder (SAE) was proposed to improve the accuracy of prediction. Srichan et al. [5] utilized features derived from multiple photonic bands near-infrared (mbNIR) and personalized medical features (PMF) to train a Shallow Dense Neural Network (SDNN), noted for its simplicity and low computational cost compared to other neural networks. In addition, Shao et al. [6] collected the obtained Raman spectrum after focusing the laser on the blood vessels in the skin, and blood glucose concentration was obtained by setting the hemoglobin concentration as the internal standard.

When it comes to applying machine learning models to blood glucose measurement, PPG has been widely utilized in biomedical applications, including blood pressure monitoring [7,8], blood oxygen measurement [9,10], and blood glucose estimation [11–14]. For example, Zhang et al. [11] used a smartphone camera to obtain PPG signals. They proposed an algorithm to eliminate baseline drift, divided the blood glucose concentration into three levels, and used features extracted by Gaussian fitting curves to train a machine-learning model. Similarly, Islam et al. [12] acquired PPG signals using a smartphone, focusing on quantitative predictions by extracting features such as the peak of the PPG signal and the time difference between diastolic and systolic blood pressure. PLSR proved effective for quantitative blood glucose prediction. Rachim and

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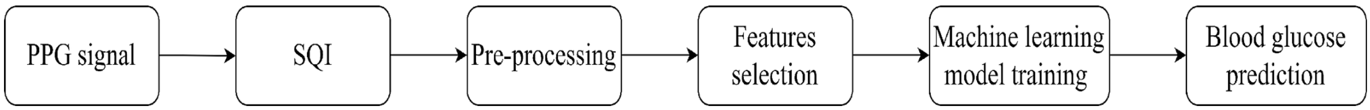

Fig. 1. Experimental Procedure for Blood Glucose Prediction.

Chung [13] developed a wearable, non-invasive blood glucose concentration monitoring system based on PPG with four light wavelengths. Before machine learning training, the processed data were categorized into three datasets—visible light, near-infrared, and composite. Models trained using composite datasets achieved the best results. Susana et al. [14] also studied the training speed and prediction accuracy of different machine-learning models using PPG signals. The Ensemble Bagged Trees classifier outperformed other models in classifying predictions as "Normal" or "Diabetes".

Recently, in building on these applications, Tjitra et al. [15] developed a wireless, noninvasive blood pressure measurement system based on PPG signals and a Random Forest regressor. By integrating the MAX30102 sensor with machine learning techniques, their study demonstrated the feasibility of continuous, cuffless blood pressure monitoring, further highlighting the potential of PPG signals for real-time physiological assessments, including glucose monitoring. Satter et al. [16] explored the use of PPG signals for blood glucose estimation, mainly focusing on fasting blood glucose prediction. Their study employed wrist-based PPG signals and extracted features such as the AC/DC ratio and optical signal ratios from different wavelengths to enhance prediction accuracy. They demonstrated the feasibility of PPG-based blood glucose monitoring by utilizing machine learning algorithms, including RF, XGBoost, CatBoost, and LightGBM. Also, Saha et al. [17] investigated the use of PPG signals for Type II diabetes prediction, focusing on extracting relevant features to enhance classification accuracy. Their study meticulously analyzed spectral entropy, Shannon energy, and other time–frequency domain features, demonstrating their effectiveness in distinguishing between normal and diabetic individuals. Additionally, Samimi and Dajani [18] proposed a calibration-free, cuffless blood pressure estimation method using PPG, demonstrating its ability to capture physiological parameters. This method underscores the broader potential of PPG for health monitoring, including blood glucose prediction. Similarly, Al Fahoum et al. [19] applied advanced feature selection techniques to PPG signals for coronary artery disease detection, emphasizing the importance of optimizing relevant features while minimizing redundancy—an approach equally beneficial for glucose level prediction models. Furthermore, Josan et al. [20] employed XGBoost to analyze short-term heart rate variability (HRV) for congestive heart failure detection, demonstrating the model's robustness in physiological signal processing, including blood glucose prediction. Their study reinforced the effectiveness of XGBoost in handling physiological signals, making it a promising choice for glucose level estimation based on PPG signals.

In this study, a commercial PPG device is used to extract signals, followed by SQI, filtering, and baseline elimination to remove the effects of noise and motion artifacts. This study extracts 36 features from PPG signals and evaluates their interrelationships using a correlation matrix, integrating both time and frequency domains to provide a comprehensive analysis. Additionally, using four distinct datasets allows for a robust evaluation of model generalization. This study extracts 36 features from PPG signals and analyzes their interrelationships using a correlation matrix. Unlike previous studies, it considers both time and frequency domains for feature extraction. Finally, 9 of the 36 features from a single light source are chosen as inputs into RF and XGBoost for predicting blood glucose levels. Four datasets— "Red," "Infrared," "Composite" ("Red + Infrared"), and "Modified Composite"—are further investigated as inputs for machine learning.

2. Methodology

2.1. Device of photoplethysmography

There are many studies of PPG that use different methods to obtain the signal of PPG. Mazzu-Nascimento et al. [21] and Zhang et al. [22] used the video recording function of smartphones to get the PPG signal. They put a fingertip on the smartphone's camera and record a video period. Rachim and Chung [13] and Gupta et al. [23] obtained PPG signals from the PPG board they designed. Like Kavsaoğlu et al. [9] and Yen et al. [24], in this study, a commercial PPG Board (MAX86150 Evaluation System, Maxim Integrated, San Jose, CA, USA) is used as the system to acquire the PPG signals.

The MAX86150 is an integrated optical biosensor module combining a photodetector, analog-to-digital converter (ADC), and two LEDs (red and one infrared). The sensor operates at a sampling rate of up to 200 samples per second and can accurately detect blood volume changes. Using dual-wavelength LEDs allows for better differentiation between oxygenated and deoxygenated blood, improving the quality of the PPG signal and enhancing the potential for non-invasive measurements such as blood glucose estimation.

The visible light (660 nm) and infrared light (880 nm) have distinct penetration depths and absorption properties. Visible light primarily interacts with superficial tissues and capillaries, whereas infrared light penetrates deeper tissues and subcutaneous blood vessels. This study utilizes these characteristics to extract complementary physiological information from the two wavelengths, leveraging their unique advantages to capture a broader range of physiological signals.

Fig. 1 illustrates the experimental workflow, starting from PPG signal acquisition to the final blood glucose prediction. The process includes SQI, pre-processing to remove noise, feature selection, and machine learning model training, demonstrating the structured approach of this study.

2.2. Signal quality index calculation

To confirm the experimental signals are not influenced by motion artifacts, Venkat et al. [10] proposed an (Kiruthiga et al., 2018)SQI to screen the good quality of PPG signals. SQI contains five signal quality parameters, namely relative location of PPG signal peak (RLP), relative location of PPG signal valley (RLV), relative peak valley index (RPVI), amplitude of PPG cycle (AP), and power spectral density index (PSDI). After getting five parameters, SQI [10] can be obtained by summing and averaging. It is noted that the signals with SQI greater than 0.5 are used to do subsequent signal preprocessing, feature extraction, and machine learning model training.

2.3. Pre-processing in PPG signal

After SQI detection, both "good" and "acceptable" categories of the PPG signals obtained from the experiment are pre-processed. This study applies a 6th-order Butterworth filter from the MATLAB function to remove frequency components above 16 Hz from the PPG signal [11]. It is noted that baseline drift is a common situation in PPG signals, and it can cause distortion in the analysis of the PPG signal for each individual cardiac cycle. Therefore, Zhang et al. [11] proposed a fitting-based sliding window (FSW) algorithm to overcome the problem of baseline drift. First, the MATLAB function stores the valley value and position in the filtered PPG signal and stores them in an array, naming the valley position as X and the valley value as Y. Then, a cubic spline function is imported to fit a line through all X and Y values. This line is considered the baseline of the PPG signal. Finally, subtracting the baseline from the PPG signal yields a baseline-removed PPG signal [11].

The chosen SQI threshold of 0.75 is based on recommendations from the literature and aims to ensure a balance between data quality and quantity. While the division between "good" and "acceptable" signals results in an approximate 1:1 ratio, this is coincidental and not the result of intentional design. Potential methods to improve signal quality and increase the proportion of "good" signals include enhancing hardware design, applying more advanced noise reduction algorithms, and optimizing measurement conditions to reduce external interference and motion artifacts.

The effectiveness of the FSW algorithm is critical for ensuring the stability of the DC component in PPG signals. By removing baseline drift, the DC component better reflects slow-changing physiological features such as tissue and blood content, reducing analysis bias caused by environmental or motion artifacts. Additionally, trend removal through FSW also indirectly improves the quality of the AC component, ensuring high-frequency variations associated with cardiac cycles are not distorted. These enhancements contribute to more consistent and reliable feature extraction, positively impacting the accuracy of machine-learning models used for blood glucose level prediction.

2.4. Feature extraction

In this study, the rationale for selecting specific features is grounded in their capability to capture key characteristics of the PPG signal that are potentially related to blood glucose concentration. Recent studies have demonstrated that advanced feature extraction and selection techniques, such as Empirical Mode Decomposition (EMD), can significantly improve the accuracy of non-invasive glucose monitoring using PPG signals [25]. By decomposing PPG signals into intrinsic mode functions (IMFs), these approaches enhance model interpretability and robustness, leading to more reliable blood glucose prediction. For instance, Power Spectral Density (PSD) was chosen for its ability to represent the energy distribution across different frequency bands, providing insights into underlying physiological processes. The PSD was calculated by applying a Fast Fourier Transform (FFT) to the segmented PPG signal, yielding the frequency-domain representation necessary for analysis. Specifically, we selected the peak frequency component from the FFT to represent the AC term of the PPG signal rather than the RMS value after DC removal. This choice was made due to its advantages in capturing high-amplitude pulsatile features and providing detailed frequency-domain information, which is critical for analyzing the relationship between blood glucose levels and PPG signals. Furthermore, the FFT's capability to suppress high-frequency noise ensures that the AC component accurately reflects variations induced by blood flow changes.

Kaiser-Teager Energy (KTE) was selected to emphasize instantaneous energy variations within the signal, which is useful for detecting rapid changes in blood flow. Time-domain features such as peak height and slope were extracted based on their known relationships with cardiovascular dynamics, which may affect blood glucose levels. Additionally, the optical density (OD) and PPG's DC/AC terms were included due to their potential to reflect tissue light absorption changes, an indirect measure of blood composition.

2.4.1. Power spectral density (PSD)

To clearly know about the energy distribution in different frequency components, Gupta's method [23] is chosen to compute PSD because of its fast computation. Using Gupta's method, the PPG signal is divided into multiple sample points based on Sampling frequency. For each sample point k, the Fourier transform $A_k(n)$ of these sequences can be calculated by Eq.(1), and then the modified periodogram can be calculated as shown in Eq.(2) as

$$A_k(n) = \frac{1}{L}\sum_{m=0}^{L-1} s_k(m)W(m)e^{\frac{-2\pi jmn}{L}} \tag{1}$$

$$I_k(f_n) = \frac{L}{U}|A_k(n)|^2 \tag{2}$$

where $s_k(m)$ is a sample from a stationary, second-order stochastic sequence, $W(m)$ stands for a window of data, and $i = \sqrt{-1}$. As a result, $\widehat{P}$ representing PSD is calculated in Eq. (3) as

$$\widehat{P}(f_n) = \frac{1}{K}\sum_{k=1}^{K} I_k(f_n) \tag{3}$$

where

$$f_n = \frac{n}{L}, U = \frac{1}{L}\sum_{m=0}^{L-1} W^2(m) \tag{4}$$

Some statistical properties, including mean, skewness, variance, and kurtosis of PSD, are used as the features in this study.

2.4.2. Kaiser-Teager energy (KTE)

KTE is a method to analyze the energy profile of the PPG signal. The equation of KTE calculation is expressed in Eq. (5) as [23]

$$\Phi[s(n)] = s^2(n) - s(n+1)s(n-1) \tag{5}$$

where s represents the PPG signal value and n represents the sample point of the PPG signal. Additionally, some statistical properties, including mean, skewness, variance, kurtosis, standard deviation, and the area under the curve of KTE, are used as the features.

2.4.3. Fast Fourier transformation

FFT is an efficient algorithm used to compute the Discrete Fourier Transform of a sequence or signal, and it is calculated in Eq. (6) as

$$F(k) = \sum_{j=1}^{n} f(j)W_n^{(j-1)(k-1)} \tag{6}$$

and

$$W_n = e^{(-2\pi i)/n} \tag{7}$$

where f stands for the original PPG signal and W_n stands for the weighting function, which is calculated as Eq. (7). Again, some statistical properties, including skewness, variance, and kurtosis of FFT, are used as the features [23].

2.4.4. Optical density (OD) and PPG's DC/AC term

Rachim and Chung [13] mentioned that the phenomenon of light attenuation in tissue can be known from the Beer-Lambert law, and its calculation is shown in Eq. (8) as

$$OD = log\left(\frac{I_0}{I}\right) \tag{8}$$

where I_0 is the intensity of light coming out of the light source and I is the intensity of light received by the detector. In the PPG system, I_0 is the amplitude of the PPG signal, and it contains the DC and AC terms. The FFT is used to get the DC term and the AC term of the PPG signal. Eq. (6) shows that the DC term is the value when n = 1. In this study, AC terms are calculated by searching the maximum peak in the curve from n = 2 to n = k. In addition, the difference in optical density can be obtained by PPG's AC and DC terms as [13]

$$\Delta OD = log\left(1 + \frac{\Delta PPG_{AC}}{PPG_{DC}}\right) \tag{9}$$

where ΔOD represents the difference in optical density, ΔPPGAC represents the difference in the AC term of the PPG signal, and PPG_{DC} represents the DC term of the PPG signal.

Table 1
Time-domain features [26].

Item Number	Indicators	Feature name
1	Height	h_1
2	Time	t_1
		t_2
		$t_1 + t_2$
		t_1-t_2
		t_1/t_2
3	Area	s_1
		s_2
		$s_1 + s_2$
		$s_1/(s_1 + s_2)$
		$s_2/(s_1 + s_2)$
		s_1/s_2
4	Slope	p_1
		p_2

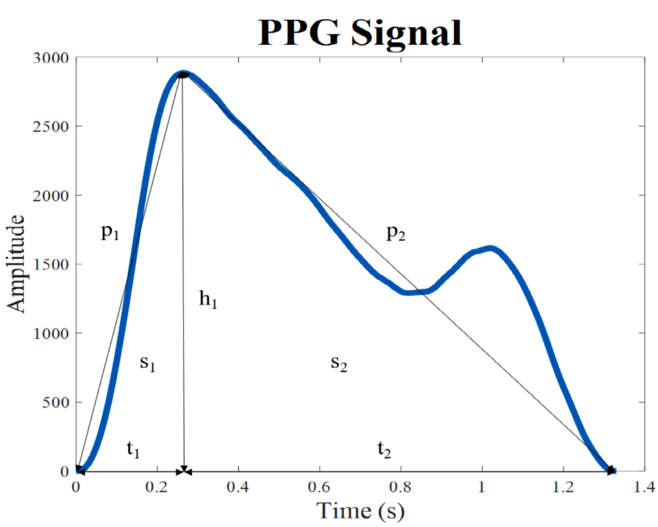

Fig. 2. Extracted Features of the PPG Signal Corresponding to Table 1.

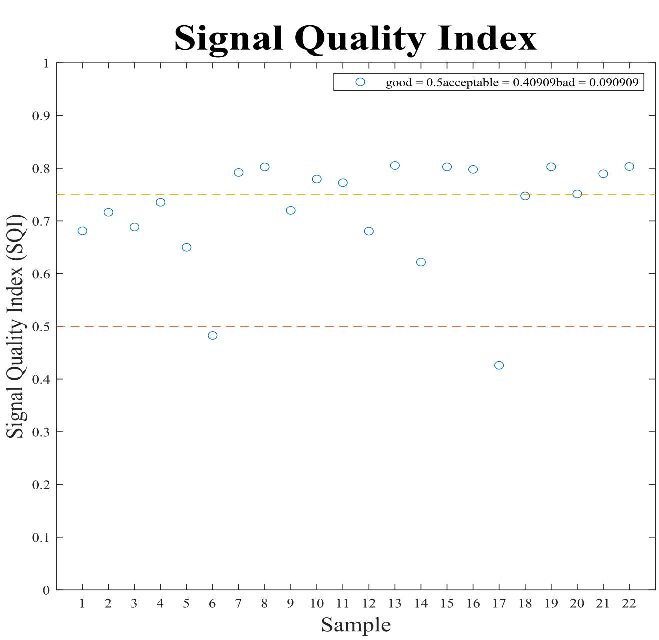

Fig. 3. SQI of PPG Signals.

2.4.5. Time-domain features

Nie et al. [26] described the characteristics of time-domain signals, mainly for the characteristics of PPG signals as features. The time-domain features of PPG signals contain the height of the peak, the time of the rising and falling branches of the waveform, and the slope of the rising and falling branches of the waveform. Also, there are some corresponding combinations considered to be the time-domain features. Table 1 illustrates the details of time-domain features and the related calculations. The other details can be found in [26].

Fig. 2 illustrates the various features extracted from the PPG signal, such as the peak amplitude (p1, p2), slope (s1, s2), time intervals (t1, t2), and height (h1). These features serve as inputs for the feature selection process, which is critical for accurate blood glucose prediction.

2.4.6. Gaussian fitting curve feature

Zhang et al. [11] studied the relationship between the features obtained by Gaussian fitting and the different levels of blood glucose concentration. In this study, the PPG signal was divided into multiple sub-signals, and then the two-term Gaussian fitting curve was used to fit the PPG sub-signals. The coefficients of Gaussian fitting are H, N, and W, respectively, representing the height of the Gaussian curve, the position of the peak, and the width at half maximum [11]. Therefore, the features of the two Gaussian fittings are H_1, H_2, N_1, N_2, W_1, and W_2, respectively, and they represent the coefficients of the two terms of the Gaussian fit.

2.5. Machine learning model

In recent years, many papers have mentioned the use of machine learning models to predict blood glucose concentrations, leveraging techniques such as Principal Component Analysis (PCA), Partial Least Squares (PLS) regression, Support Vector Regression (SVR), Random Forest (RF), and eXtreme Gradient Boosting (XGBoost). Among these, RF and XGBoost are recognized for their higher prediction accuracy due to their decision-tree-based learning architecture, which effectively captures non-linear relationships and complex feature interactions [27]. These models have been extensively applied in biomedical signal analysis and have demonstrated robust performance across various physiological data applications.

RF is a machine-learning model capable of handling both regression and classification tasks. It constructs an ensemble of decision trees during training, with each tree trained on a random subset of the original training data. This process, known as bootstrapping, reduces overfitting and enhances the model's generalizability [27]. One notable strength of RF is its ability to manage non-linear data and provide insights into feature importance, helping identify which features contribute most significantly to predictions.

In addition, XGBoost is a highly efficient algorithm in the gradient-boosting family [28]. It builds an ensemble of weak prediction models, typically decision trees, in a sequential manner, where each new tree aims to correct the errors of the previous ones. This iterative process optimizes learning and results in high prediction accuracy. XGBoost's use of regularization techniques, such as L1 (lasso) and L2 (ridge) penalties, helps prevent overfitting and improves the model's generalization to new data. Furthermore, XGBoost supports parallel processing, significantly speeding up the training process and making it well-suited for large-scale data analysis.

In conclusion, this study applies RF and XGBoost models, combined with the previously collected features, to predict blood glucose levels. These models were chosen due to their proven capability to handle complex, non-linear data effectively and their widespread application in similar biomedical signal prediction tasks.

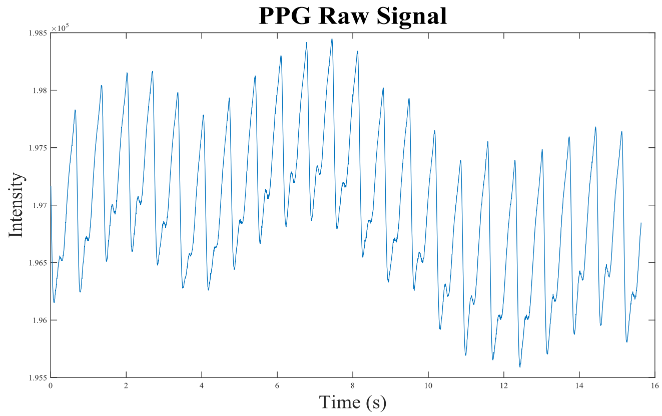

Fig. 4. Raw signal from the PPG device.

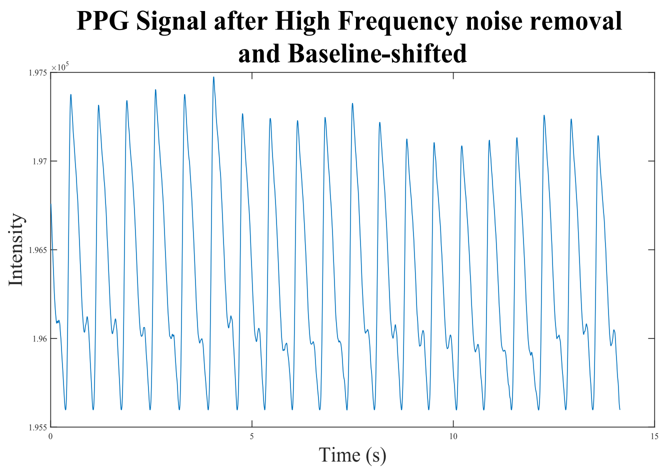

Fig. 5. Signal after high-frequency noise removal and baseline shifted.

3. Experimental setup and feature selection

3.1. Experimental setup

The PPG commercial module used in this study is the MAX86150 Evaluation System (Maxim Integrated, San Jose, CA, USA). The module has two different wavelengths of light source: the visible light of 660 nm and the near-infrared of 880 nm. In this research, a 23-year-old male participant with no physiological diseases was asked to undergo testing during three different periods of time. It is noted that PPG signals were extracted during each 10-second test period. First, the participant was instructed to perform the test in a fasting state. Then, the participant was asked to consume a meal and set a one-hour timer from the start of eating. After one hour, the second test was conducted, followed by a third test two hours post-meal. Additionally, the participant underwent an invasive blood glucose measurement after each test. The commercial invasive glucose measurement device (Accu-Chek Instant blood glucose meter) was used for this purpose. To ensure consistent signal quality and minimize external interferences, all tests were conducted with the room lights turned off.

3.2. Signal quality index

There are a lot of factors that could affect the purity of the signal, and motion artifact is one of the facts. SQI is defined to confirm the signal is not affected by motion artifacts. Fig. 3 shows the data of one period of experiments in one day of the SQI of the PPG signals from the same tester. The SQI value between 0.75 and 1 is categorized as “good”. Also, the SQI value between 0.75 and 0.5 is classified as “acceptable”. The PPG signal with its SQI value less than 0.5 would be eliminated. In Fig. 3, the signals of the “good” and “acceptable” categories are used as the data for machine learning prediction.

Table 2
Assigned Number and Name of Feature.

Number	Name	Number	Name	Number	Name
1	PSD	13	Variance FFT	25	$s_1 + s_2$
2	Kurtosis PSD	14	PPG DC term	26	$s_1/(s_1 + s_2)$
3	Variance PSD	15	PPG AC term	27	$s_2/(s_1 + s_2)$
4	Mean PSD	16	ΔOD	28	s_1/s_2
5	Kurtosis KTE	17	h_1	29	p_1
6	Variance KTE	18	t_1	30	p_2
7	Mean KTE	19	t_2	31	H_1
8	Skewness KTE	20	$t_1 + t_2$	32	H_2
9	AUC KTE	21	t_1-t_2	33	n_1
10	Standard deviation KTE	22	t_1/t_2	34	n_2
11	Kurtosis FFT	23	s_1	35	W_1
12	Skewness FFT	24	s_2	36	W_2

3.3. Signal processing in noise removal and baseline shifted

This study has two pre-processing steps in noise removal and a shifted baseline. The raw signal measured by the PPG device is shown in Fig. 4. To remove high-frequency noise, a 6th-order pass Butterworth filter with 16 Hz of cutoff frequency is used in the first step [11]. In addition, baseline drifting is a common problem in PPG signal, as shown in Fig. 4. To overcome baseline drifting, the FSW algorithm proposed by Zhang et al. [11] is implemented. Finally, the result of the PPG signal after going through noise filtering and the FSW algorithm is shown in Fig. 5. The FSW algorithm has effectively solved the baseline drifting problem in the PPG signal.

3.4. Feature selection

After signal processing in raw PPG signal, the feature extractions introduced in Section 2 are applied, and all the assigned number and name features are sorted out in Table 2. In addition, since there are two light sources designed in a commercial PPG system, namely 660 nm and 880 nm, the feature extractions obtained from these two light sources are all used for studying correlation matrix analysis and machine learning.

To select appropriate features as the input data of machine learning, exploring the correlation between each feature is important. First, use a correlation matrix based on the Pearson correlation coefficient from Python to calculate the correlation between the features. Then, the correlation matrix is visualized using the heatmap from MATLAB. As a result, Fig. 6 illustrates the result of the heatmap of the correlation matrix. It is noted that the dark color of the grid means there is a high correlation between two features, and the light color of the grid means the correlation between two features is low. Furthermore, if the machine learning input data have a high correlation with each other, the weights are dispersed. Thus, the weights of different features with a high correlation with the predicted target would be assigned less. As a result, it would lead to a decrease in prediction accuracy [3]. For this case, to eliminate the high correlation feature with other features is critical for enhancing the accuracy in prediction by using machining learning.

According to Fig. 6(a) for the red light, it is easy to observe that there is a high correlation among Features 1, 3, 4, 13, and 14. Similarly, there is a high correlation among Features 15, 16, 17, 23, 24, 25, 29, 30, 31, and 32. In addition, there is a high correlation among Features 5 and 8, 11 and 12, as well as 26 and 28. Again, according to Fig. 6(b) for the

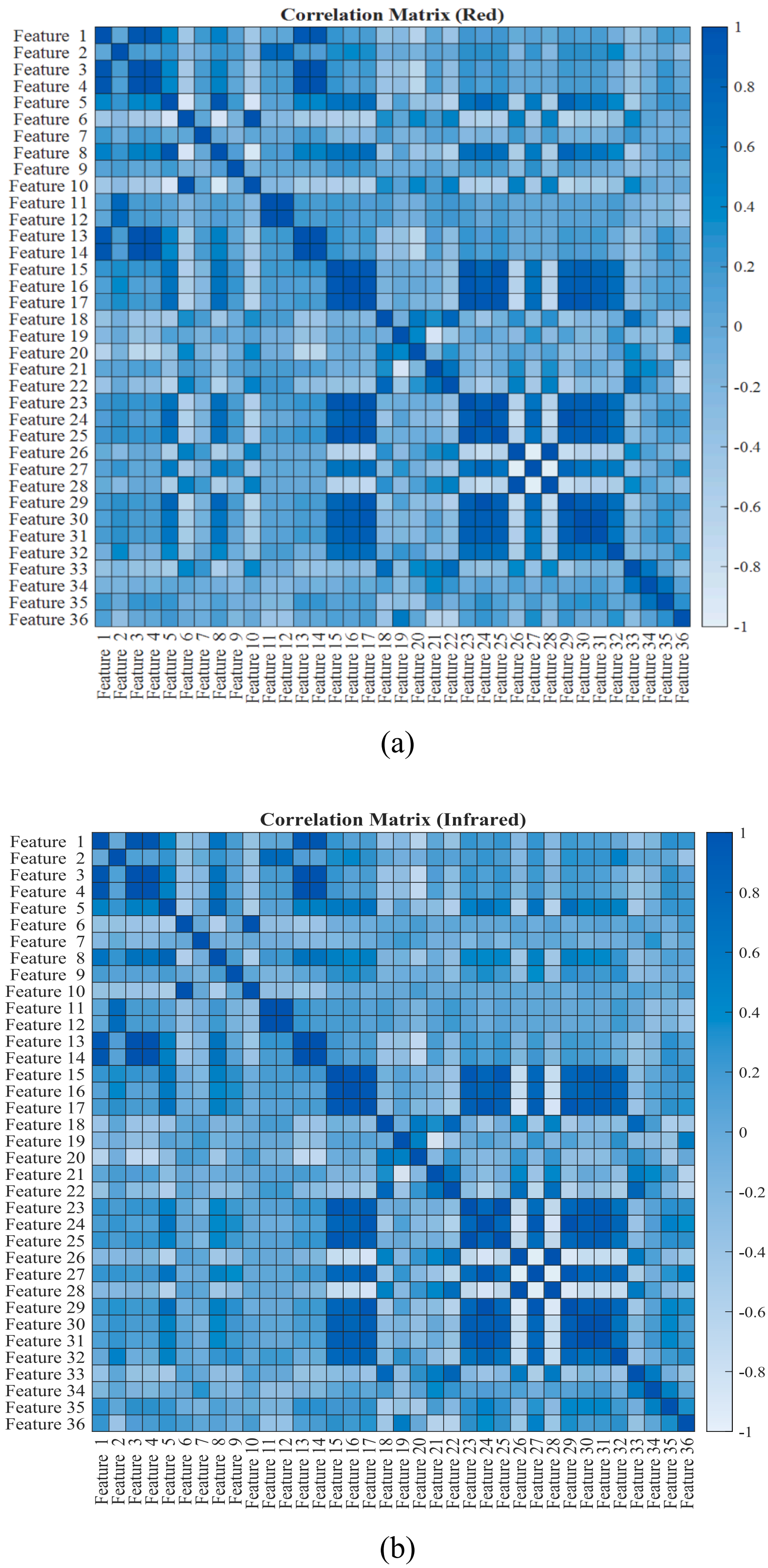

Fig. 6. Feature correlation heatmap for (a) Red light; and (b) Infrared light. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

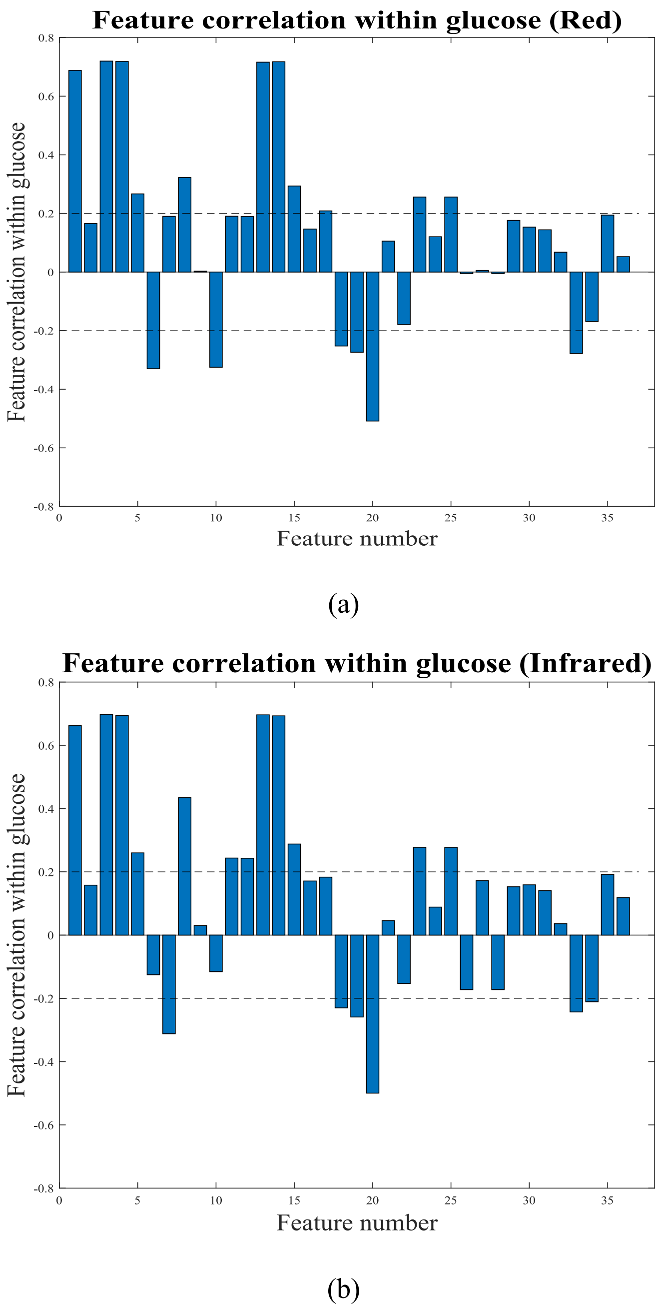

Fig. 7. Correlation between glucose level for (a) Features of red light; and (b) Features of infrared light. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

infrared light, Features 1, 3, 4, 13, and 14 have a high correlation. Also, Features 15, 16, 17, 23, 24, 25, 29, 30, 31, and 32 have a high correlation. In addition, there is a high correlation among Features 5 and 8, 11 and 12, as well as 26 and 28. It is found that the correlation matrices in Fig. 6(a) and 6(b) for the red and infrared lights are similar. It is noted that the high collinearity among features can reduce the accuracy of machine learning models in predictions; therefore, excluding highly collinear features in improving the accuracy of predictions is needed.

In addition to considering the correlation between features, the correlation between features and blood glucose level is also very important. Fig. 7(a) for red light shows the correlation between features and glucose value, and the dotted line setting between 0.2 and − 0.2 is designed for searching the features that have a correlation with the predicted target. It is found that Features 1, 3, 4, 5, 8, 13, 14, 15, 17, 23,

Table 3
Number and name of the feature in the composite dataset.

Red		Infrared	
Number	name	Number	Name
1	Mean KTE	10	Mean KTE
2	Skewness KTE	11	Skewness KTE
3	PPG DC term	12	PPG DC term
4	PPG AC term	13	PPG AC term
5	t_1	14	t_1
6	t_2	15	t_2
7	$t_1 + t_2$	16	$t_1 + t_2$
8	n_1	17	n_1
9	n_2	18	n_2

and 25 are the features having a correlation of more than 0.2 with the blood glucose level. Again, Features 6, 10, 18, 19, 20, and 33 correlate less than −0.2 with the blood glucose level. As a result, the features whose correlation with blood glucose is greater than 0.2 and less than −0.2 are listed as potential candidate features for machining learning. Subsequently, their collinearity between features is carefully considered, and the final features selected as inputs of machine learning are Features 7, 8, 14, 15, 18, 19, 20, 33, and 34, respectively, for a red light.

Similarly, in Fig. 7(b) for infrared light, Features 1, 3, 4, 5, 8, 11, 12, 13, 14, 15, 23, and 25 correlate higher than 0.2 with the predicted target. Features 7, 18, 19, 20, and 33 are the features having a correlation of less than 0.2 with blood glucose concentration. Subsequently, their collinearity between features is again carefully considered, and the final features selected as inputs of machine learning are Features 7, 8, 14, 15, 18, 19, 20, 33, and 34, respectively, for infrared light. Finally, in the training phase, there are three datasets investigated in this study, namely "Red", "Infrared", and "Composite" ("Red + Infrared") datasets.

3.5. Modify the composite dataset

Rachim and Chung [13] have mentioned that the dataset obtained from two light sources, visible light and infrared light, has better predicted results than the dataset obtained only from a single light source. Accordingly, it is possible that the collinearity of the features in the composite dataset [13] needs to be further investigated as the modified composite dataset in this study. Therefore, the feature's number and name of the composite dataset are illustrated in Table 3. It is found that Features 1–9 are selected from a red band and Features 10–18 are selected from an infrared band. The correlation heatmap is shown in Fig. 8, and it can be clearly observed that the same type of features from different light sources has a high correlation, as indicated by the yellow and white diagonal lines in the diagram. However, only Features 1 and 10 don't exhibit a high correlation across themselves, although the two features have the same attributes," Mean KTE", with different light sources as indicated by the red circles in Fig. 8. Accordingly, in this case, modifying the composite dataset was necessary to account for their collinearity. By comparing the correlation between the same type of features and the blood glucose level from two light sources, the features from one of the light sources that have a higher correlation with blood glucose concentration are specifically selected, while the same features from the other light source are removed.

Comparatively, under infrared light, the skewness of KTE (Feature 11) and n2 (Feature 18) show a stronger correlation with blood glucose compared to their performance under red light (Features 2 and 9), as indicated by the dashed box in Fig. 9. Specifically, Features 11 and 18 have higher positive correlation coefficients under infrared light, suggesting that these features more effectively reflect blood glucose levels in this light condition. In contrast, Features 2 and 9 under red light exhibit lower positive correlations. These observations indicate that, under the light source of this commercial module, using infrared light for measurements might be more advantageous in capturing important features related to blood glucose. Compared to features derived from

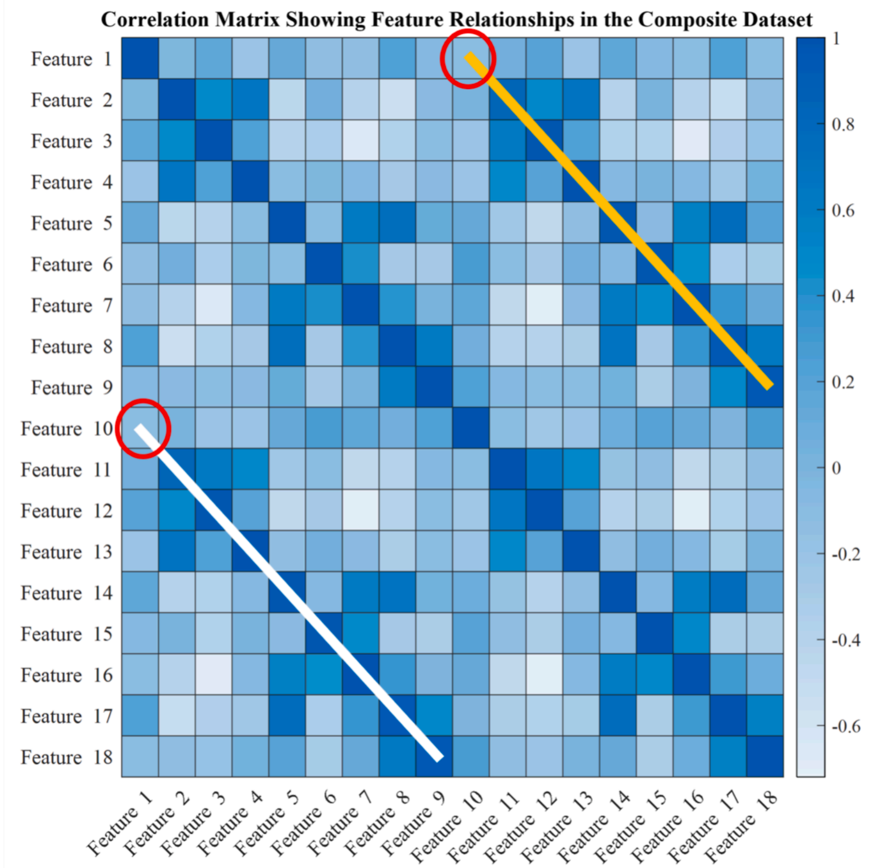

Fig. 8. Correlation Heatmap: Yellow and White Lines Indicate High Correlation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

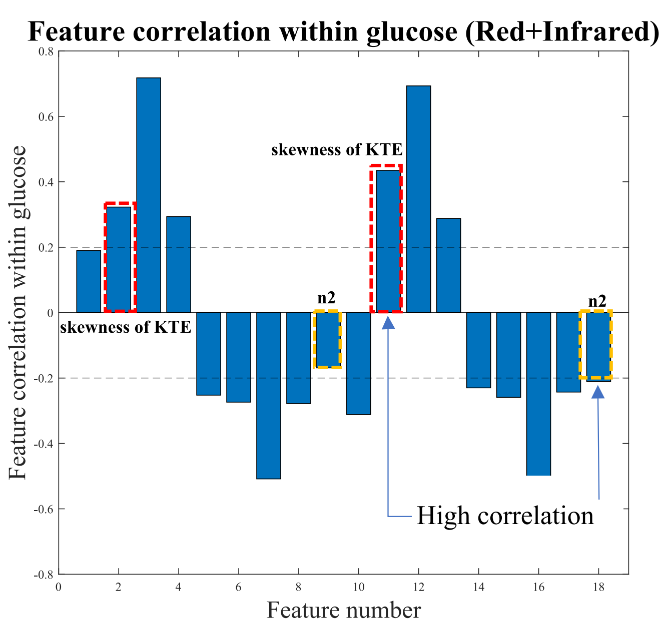

Fig. 9. Correlation between glucose and features in the composite dataset.

infrared light, those obtained from red light exhibit stronger correlations with blood glucose levels. As a result, Features 1, 3, 4, 5, 6, 7, 8, 10, 11, and 18 in Table 3 are selected explicitly as input data of the machine learning for the "Modified Composite" dataset. It is found that the number of selected features in Table 3 is much less than those in Table 2 due to the consideration of the collinearity and the correlation among the predicted targets. Once the redundant features are removed based on the collinearity analysis, the accuracy of prediction using machine learning would be increased.

4. Machine learning in prediction

This study used two machine learning models, RF and XGBoost, to predict human blood glucose concentration. There is a total of 40 data from 13 days collected from the same tester, which are divided into a training set and a testing set in an 8:2 ratio. The training set contains the data needed to be trained by the machine learning models, while the testing set is used to evaluate the performance of the trained models. Several evaluations, including MARD and R-value, are used in this study to investigate the performance of the trained models. MARD is an index used to evaluate the accuracy of blood glucose measurement systems, and its formula is shown in Eq. (10) and Eq. (11) as

$$ARD_k = \frac{\left| y_{CGM}(t_k) - y_{ref}(t_k) \right|}{y_{ref}(t_k)} \tag{10}$$

and

$$MARD = \frac{1}{N_{ref}} \sum_{k=1}^{N_{ref}} ARD_k \tag{11}$$

where y_{CGM} represents the glucose value predicted by the machine learning model, y_{ref} represents the blood glucose value obtained by a commercial glucose meter, and N_{ref} represents the number of testing

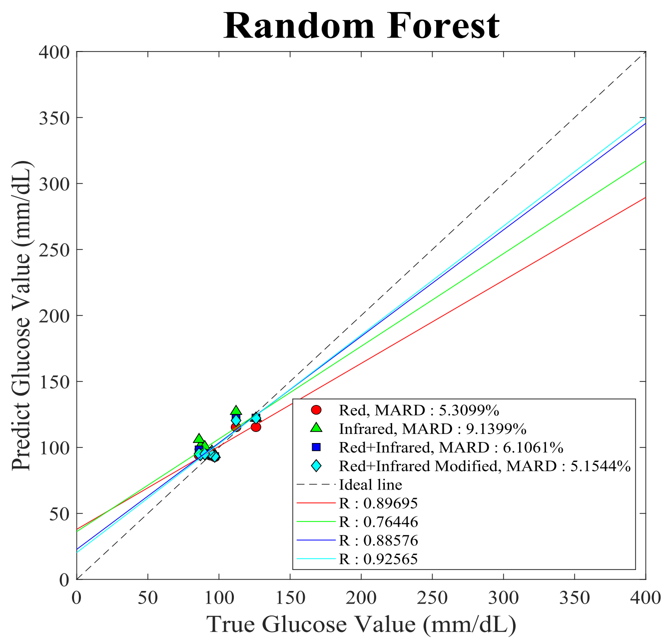

Fig. 10. Training results of the RF model for four datasets.

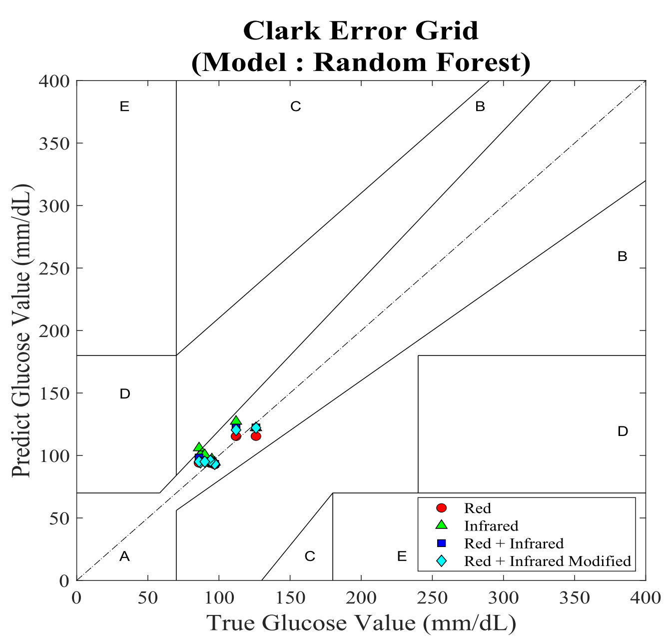

Fig. 11. Clark error grid analysis of RF model for four datasets.

data.

4.1. Predicted result of four classes of dataset

In this section, four classes of the dataset, namely "Red", "Infrared", "Composite", and "Modified Composite" are used to predict the blood glucose level using RF and XGBoost machining learning models. "Composite" contains 18 features total, 9 features from the red source and 9 features from the infrared source without collinearity analysis, while "Modified Composite" contains 10 features total, 7 features from the red source, and 3 features from the infrared source in eliminating the same types of features because of the feature collinearity. Fig. 10 shows the predicted results of the RF model based on four datasets. It can be observed that after modifying the composite dataset, the predicted outcome is the best among the four datasets. The MARD of the "Modified Composite" dataset is 5.15 % and the R-value is 0.93. Also, the Clark error grid of the RF model with four datasets is shown in Fig. 11. It can be seen from the figure that all points fall in Zone A, which means that using four kinds of datasets to respectively train the RF model and the predicted blood glucose level is all within the clinically accurate range.

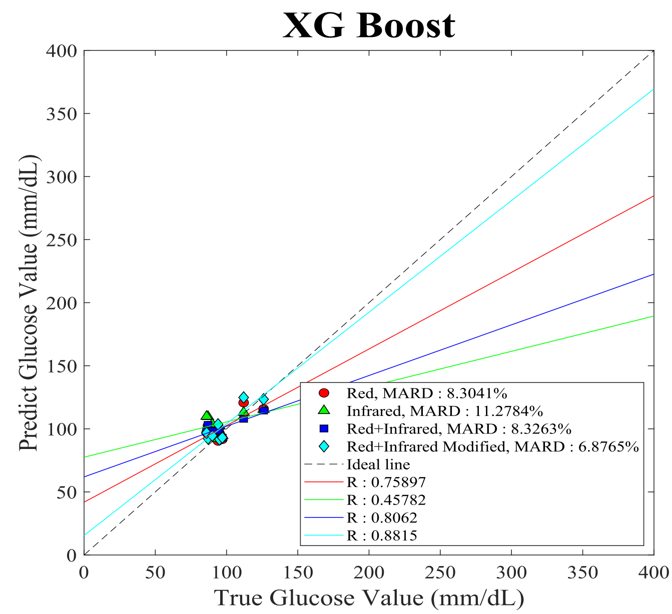

Fig. 12. Training results of the XG Boost model for four datasets.

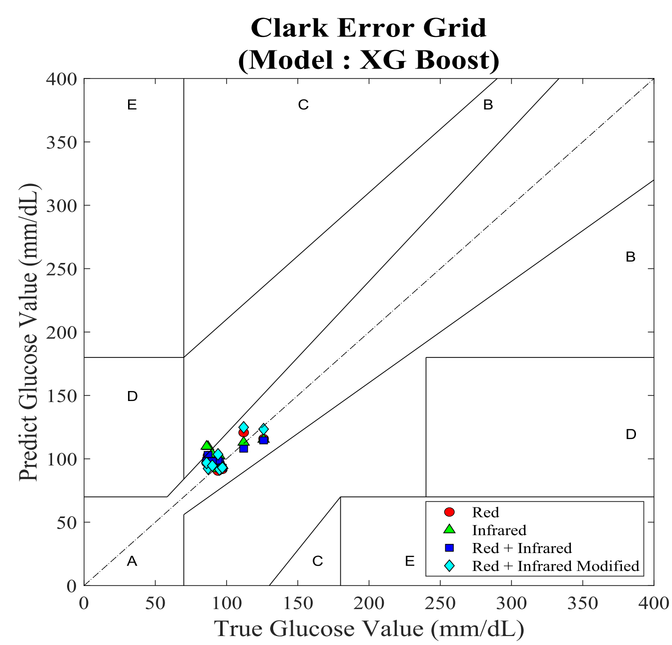

Fig. 13. Clark error grid analysis of XGBoost model for four datasets.

Fig. 12 shows the predicted results by the XGBoost model based on four datasets. It can be clearly observed that using the modified composite dataset to train the XG Boost model can get the best prediction results, and its MARD is 6.88 % and R-value is 0.88, respectively. This illustrates that due to removing the collinearity and selecting a high correlation within blood glucose level, the negative factors that caused the model to reduce the prediction accuracy were all excluded. Again, the Clark error grid of the XGBoost with four datasets is shown in Fig. 13. Although some scattering points of the dataset from infrared are falling

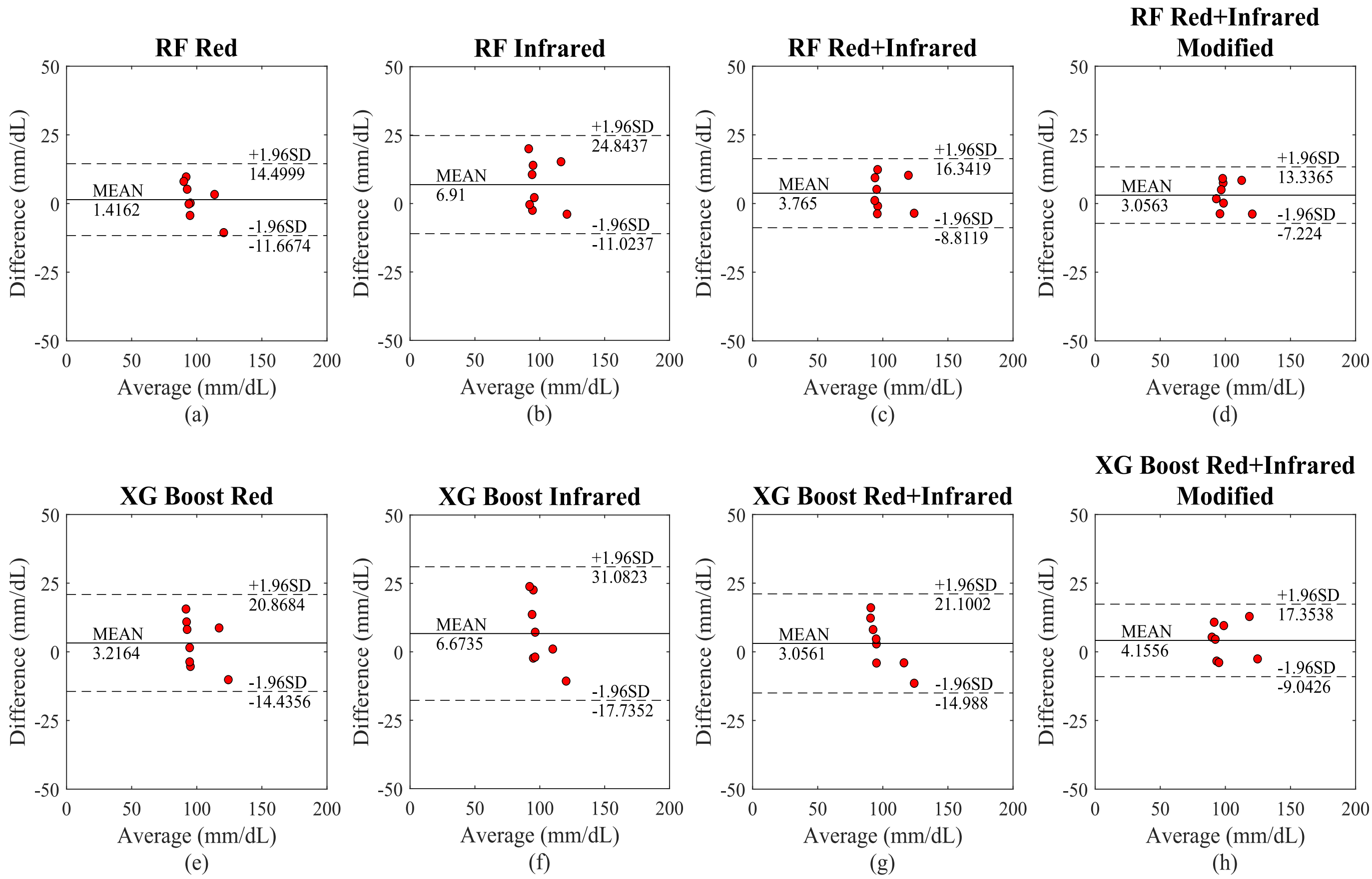

Fig. 14. Bland-Altman plot analysis of (a) – (d) RF model; and (e) – (h) XGBoost model for four datasets.

Table 4
Proposed system in comparison with previous methods.

Research Groups	Accuracy			Experimental Setup	
	R-value	MARD (%)	Zone A in CGE (%)	Participants	Measurement Device
Gupta et al. [23]	0.94	–	96.15	26	Custom-built PPG system with red, green, and IR LEDs
Hina and Saadeh [29]	0.937	7.62	95	200	Single-wavelength near-infrared (NIR) PPG system
Rachim and Chung [13]	0.86	–	100	12	Wearable wristband with visible and near-infrared biosensor
Ramasahayam et al. [30]	0.9	–	95.38	100	Near-infrared (NIR) PPG system on FPGA
Our method	0.93	5.15	100	1	Commercial MAX86150 PPG module

in Zone B, which is out of the range of 20 % within the reference value, the results of the other three datasets are all fall in Zone A, which means the predicted results are close to the reference value. Especially for the modified composite dataset, the scatter points are very close to the ideal dashed line in the middle, which means the modified composite dataset effectively improves the accuracy of machine learning model training.

4.2. Bland-Altman plot of four classes of dataset

The Bland-Altman plot [13,20] is a statistical graphical method often used to compare the consistency and the limits of consistency between the reference and predicted data. The graph shows the relationship between the difference value and the mean value of the reference and predicted data. Fig. 14 shows the Bland-Altman plot of two machine learning models with four categories of the dataset, namely “Red”, “Infrared”, “Composite”, and “Modified Composite” datasets. It can be observed that using the “Modified Composite” dataset as the input of both machine learning models has the smallest difference from −7.224 to 13.3365 and from −9.0426 to 17.3538 between actual blood glucose value and predicted value regarding RF and XGBoot machining learning models, respectively.

4.3. Comparisons with other literature’s results

Table 4 summarizes the performance of other literature on PPG and machine learning predictions. In overviewing the results of the machine learning prediction in this study, the best combination of our predicted results is by using the RF model with a “modified composite” dataset, in which the R-value is 0.93 and MARD is 5.15 %, and the scattering points in the Clark error grid are all falling in Zone A. It was found that the method proposed in this study is better than that of other studies.

Table 5
Number and name of the features in Reference [13].

Red		Infrared	
number	Name	Number	Name
1	PPG DC term	7	PPG DC term
2	PPG AC term	8	PPG AC term
3	ΔOD	9	ΔOD
4	Variance KTE	10	Variance KTE
5	Skewness KTE	11	Skewness KTE
6	Standard deviation KTE	12	Standard deviation KTE

Compared to previous methods, our approach demonstrates significant improvements. For instance, Gupta et al. achieved an R-value of 0.94 and 96.15 % of points in Zone A, but their MARD is not reported. Hina and Saadeh [29] reported an R-value of 0.937 with a MARD of 7.62 % and 95 % of points in Zone A, showing lower accuracy in terms of MARD compared to our method. Similarly, Rachim and Chung achieved a perfect 100 % of points in Zone A but had a lower R-value of 0.86. Ramasahayam et al. [30] reported an R-value of 0.9 and 95.38 % of points in Zone A, also falling short compared to our results.

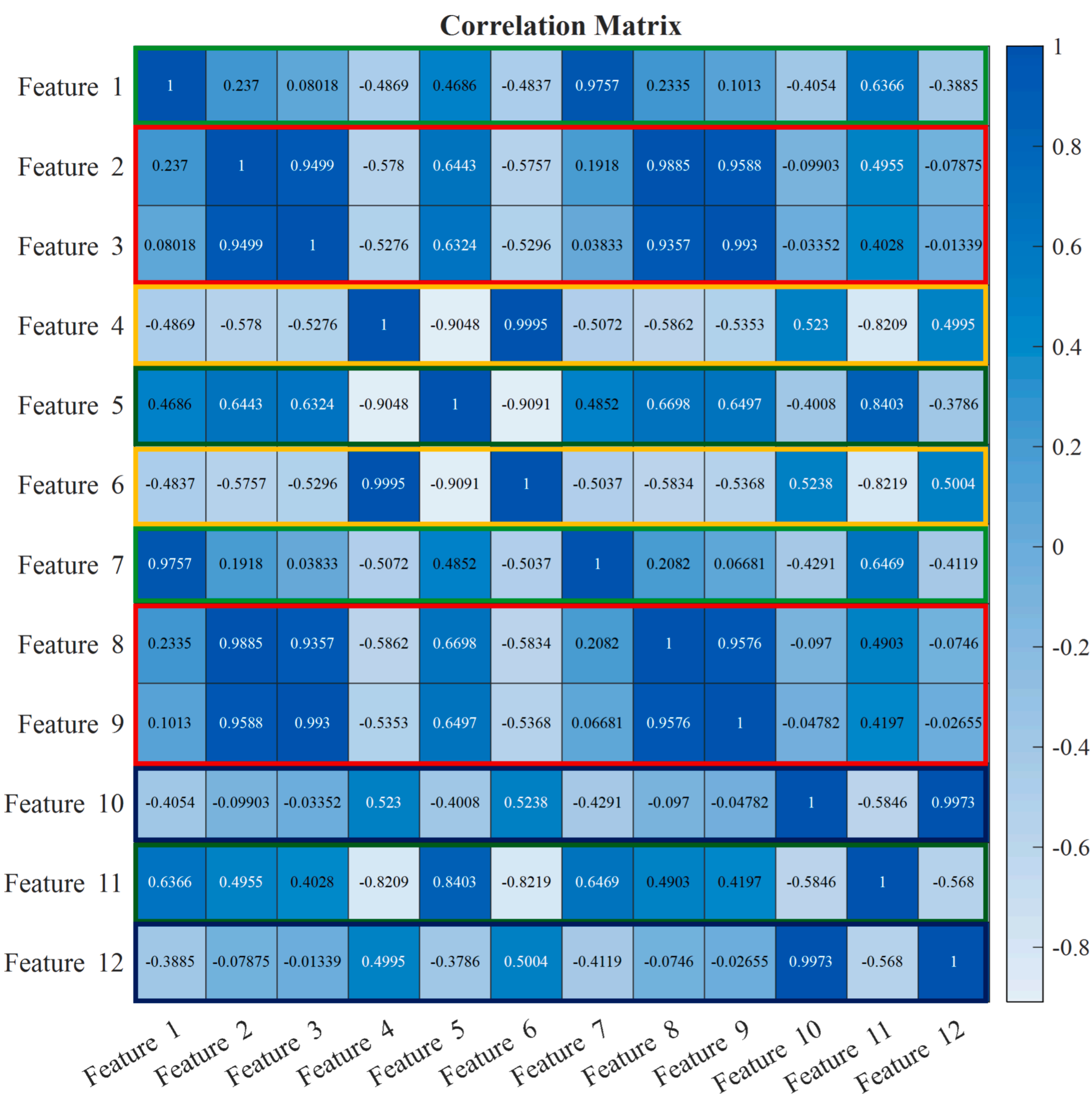

Fig. 15. Correlation Matrix of the features from Reference [13].

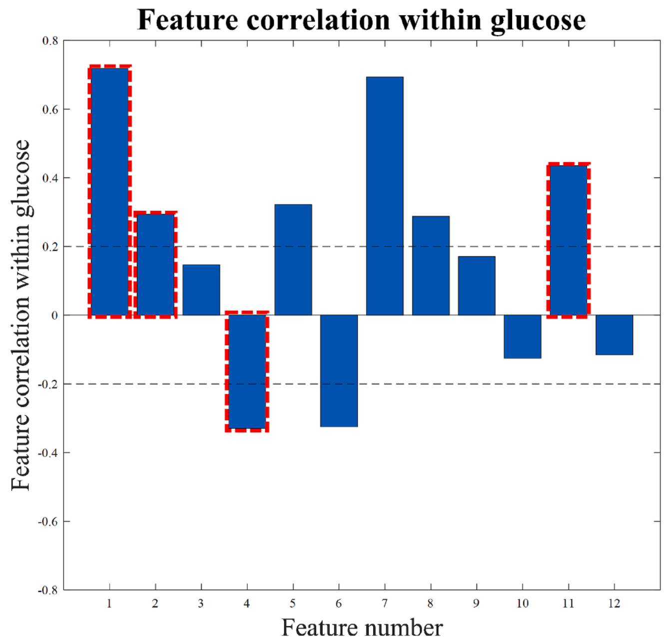

Fig. 16. Correlation between glucose and features from Reference [13].

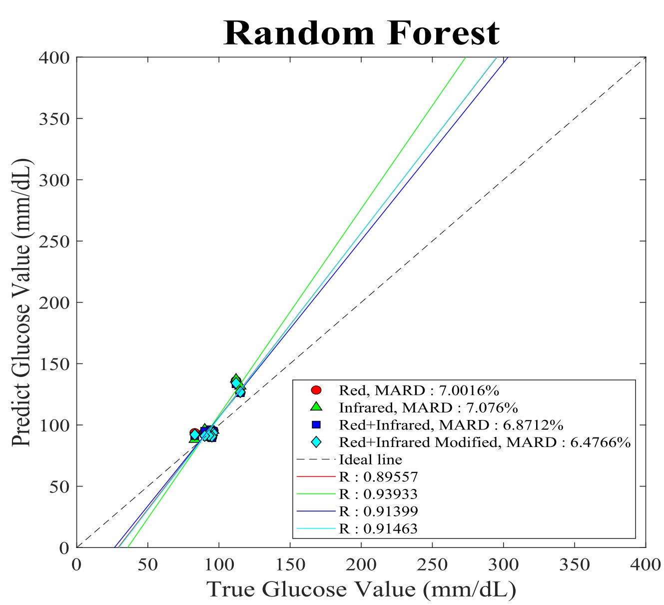

Fig. 17. Predicted result of using RF model with the features from [13].

5. Demonstration with features from a reference in [13]

5.1. Machine learning using features in prediction

Table 4 shows that the prediction results from Rachim and Chung [13] achieved 100 % in Zone A and an R-value of 0.86, demonstrating competitive performance. Consequently, their approach is chosen for further comparisons using the same original PPG dataset extracted in this study. In their work, Rachim and Chung [13] utilized 12 features to train two machine learning models: RF and XGBoost. Additionally, they employed two light sources, red and infrared, with the specific features listed in Table 5. It should be noted that the "Modified Composite" dataset in [13] was also used for further evaluation.

The correlation matrix for these features is presented in Fig. 15. Features 2, 3, 8, and 9 exhibit strong correlations, as highlighted by the red boxes in Fig. 15. Similarly, Features 4 and 6 show strong correlations, marked by yellow boxes, while Features 10 and 12 are indicated by purple boxes. Furthermore, Features 1 and 7, as well as Features 5 and 11, exhibit high correlations due to their similar properties derived from different light sources, as indicated by the green and brown boxes in Fig. 15.

Subsequently, Fig. 16 illustrates the correlation of these features with blood glucose concentration. Features 1, 2, 5, 7, 8, and 11 show a positive correlation higher than 0.2 with the predicted target, whereas Features 4 and 6 exhibit negative correlations below −0.2. Based on the correlation matrix and their relationship with blood glucose concentration, Features 1, 2, 4, and 11 are selected for inclusion in the "Modified Composite" dataset, as indicated by the red dashed boxes in Fig. 16.

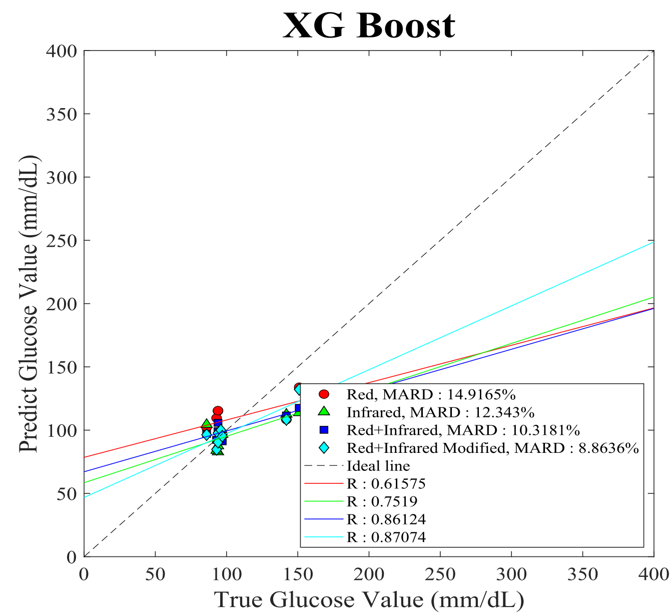

Fig. 18. Predicted result of using XG Boost model with the features from [13].

5.2. Predicted result of RF and XGBoost models

Fig. 17 presents the prediction results of the RF model using the features from Reference [13]. Among the four datasets, the "Modified Composite" dataset demonstrates the best performance, achieving a MARD of 6.48 % and an R-value of 0.91. Similarly, Fig. 18 shows the prediction results of the XGBoost model using the same features. Once again, the "Modified Composite" dataset yields the best results, with a MARD of 8.86 % and an R-value of 0.87.

When comparing the results, the RF model using the "Modified Composite" dataset with features from Reference [13] achieves the highest performance among their configurations (MARD = 6.48 %, R-value = 0.91). However, the features selected in this study outperform those from [13], as they incorporate additional time-domain features and Gaussian fitting coefficient features, which enhance the model's predictive capability.

This analysis highlights that increasing the number of features can improve prediction accuracy, provided there is no high collinearity among them. For example, this study utilizes 10 features as inputs for machine learning models, compared to the 4 features used in [13], leading to better overall performance.

6. Conclusions and discussions

This study utilized PPG signals to extract various features, including the DC and AC terms of the PPG signal, such as PSD, KTE, FFT, time-domain features, and Gaussian fitting features. All features were derived from light sources with wavelengths of 660 nm and 880 nm, respectively. These features were organized into four datasets: “Red”, “Infrared”, “Composite”, and “Modified Composite”. Two machine learning models, RF and XGBoost, were employed to train and predict blood glucose concentrations. The “Modified Composite” dataset was designed by selecting features with low collinearity and high correlation with blood glucose concentration. The results demonstrated that the “Modified Composite” dataset combined with the RF model achieved the best performance, with a MARD of 5.15 % and an R-value of 0.93.

Additionally, the 12 features referenced by Rachim and Chung [13] were tested as input for machine learning in training and testing. Comparatively, the features proposed in this study yielded superior results due to the inclusion of time-domain features and Gaussian fitting coefficient features, which enhanced the predictive performance. It is worth noting that the proposed PPG-based system for extracting glucose concentration is designed for individual use rather than as a generalized device, positioning it as a custom-made solution.

Although this study successfully extracted 36 features from PPG signals, the correlation between these features and blood glucose concentration remains moderate. This suggests the need to explore more sensitive features of the PPG signal in future research. Additionally, integrating light sources with different wavelengths may enhance the correlation between certain features and blood glucose concentration.

The findings from this study have significant implications for non-invasive glucose monitoring and other biomedical applications. The feature extraction techniques and machine learning models used here can potentially be extended to predict other physiological parameters, such as blood pressure and vascular health indicators. PPG signals are widely utilized in hemodynamic studies, and including both time-domain and frequency-domain features, as demonstrated in this study, offers a promising foundation for future applications in blood pressure prediction. For instance, incorporating systolic and diastolic time intervals into the machine learning framework may improve accuracy in blood pressure estimation, which is highly relevant for cardiovascular health monitoring.

Furthermore, this study used a single tester, which limits its generalizability. Future research should include a more extensive and diverse group of participants to examine potential inter-individual variability and its impact on the machine learning model’s performance. Addressing these variations is critical for developing a robust system that can adapt to different individuals’ physiological characteristics.

Lastly, while the results of this study emphasize the potential of the “Modified Composite” dataset and selected features, expanding the scope to incorporate additional features—potentially derived from new sensing techniques or multimodal data—could further improve predictive accuracy and broaden the system’s applicability across various healthcare domains.

CRediT authorship contribution statement

Shi-En Jian: Conceptualization, Data curation, Formal analysis, Writing – original draft. **Yu-Lung Lo:** Conceptualization, Methodology, Project administration, Supervision. **Yun-Tzu Chuang:** Validation, Writing – review & editing. **Shu-Han Kuo:** Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yu-Lung Lo reports financial support was provided by Ministry of Science and Technology, Taiwan. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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